

Automation of Yarn Tension Control with DCS Integration and Digital-Twin Predictive Maintenance

Abstract :

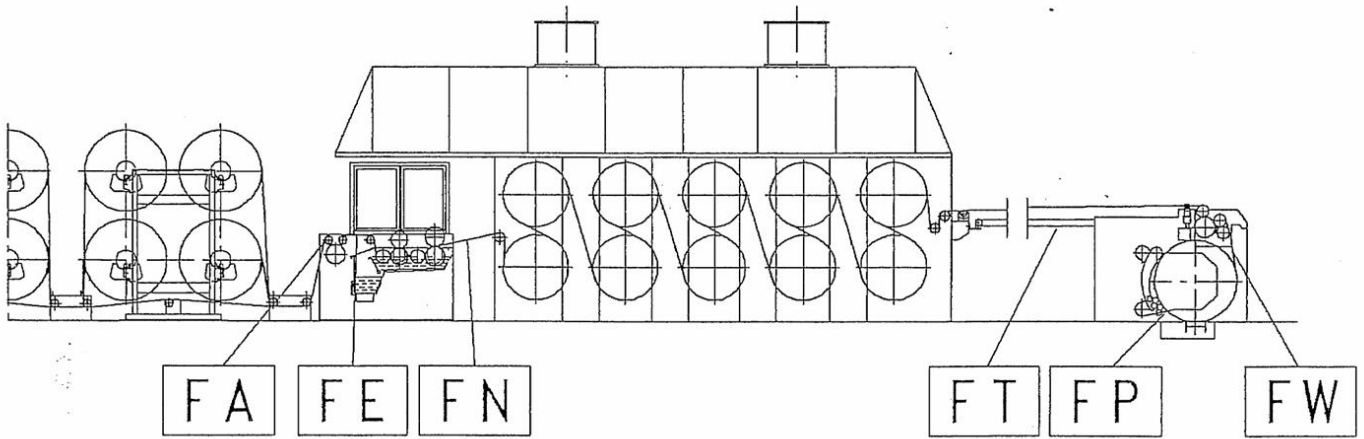
Yarn tension control is a critical aspect of the sizing process in textile manufacturing, as it directly influences yarn quality, weaving performance, and fabric appearance. Maintaining uniform yarn tension throughout the sizing machine minimizes yarn breakage, ensures even size penetration, and improves the adhesion of the sizing agent to the yarn surface. Effective tension control also reduces variations in yarn elongation and prevents defects such as uneven coating, hairiness, and distortion. Modern sizing machines employ advanced tension monitoring and feedback control systems to maintain optimal tension under varying operating conditions.

Introduction:

The sizing process is one of the most important preparatory operations in textile weaving, aimed at improving the strength, abrasion resistance, and weaving performance of warp yarns. During sizing, a protective film of sizing material is applied to the yarn surface to reduce friction, minimize hairiness, and withstand the mechanical stresses encountered during weaving. Among the various process parameters, yarn tension plays a vital role in determining the quality and efficiency of the sizing operation.

Proper yarn tension control ensures uniform stretching of the yarn, consistent penetration and adhesion of the sizing solution, and even drying across the warp sheet. Excessive tension may cause yarn elongation, weakening, or breakage, while insufficient tension can result in poor size application, uneven coating, and reduced weaving efficiency. Therefore, maintaining optimal and uniform tension throughout the sizing machine is essential for producing high-quality warp yarns.

Tension Zones :



Zone	Full Name	Location	Main function
FA	Feed area	Between beam creel and first rollers	Equalizes Yarn tension coming from the warp beams
FE	Entry Zone	Before entering sizing box	Stabilizes sheets before size pickup
FN	Main Machine	Between sizing boxes and dryers	Controls Tension during wet processing and drying
FT	Take-off Tension	Exit of dryers (Leasing section)	Stabilizes dried yarn before leasing (separation zone)
FW	Winding zone	Between draw rollers and winding	Maintains winding tension
FP	Press Zone	Weaver Beam Winder	Final Beam formation to ensure beam density and edge alignment

FA

This is arguably the most important tension zone, because it determines how the yarn sheet enters the machine.

It begins at the warp beam creel and ends at the first driven roller entering the sizing machine.

Without tension control, every beam would deliver yarn at a different tension.

If the warp sheets entered the size box without controlling the tension those would be the consequences :

- loose yarns
- tight yarns
- crossed ends
- missing ends
- uneven size penetration
- broken ends

Once yarn enters the sizing box, these problems become almost impossible to eliminate.

The sizing machine cannot fix poor incoming tension.

It only processes whatever enters it.

Effects of High FA Tension:

- excessive yarn elongation
- weak yarn
- fiber damage
- more end breaks
- poor size penetration because the yarn becomes compact
- higher warp beam pressure
- increased friction

Effects of Low FA Tension:

- slack warp sheet
- yarn overlap
- ribbon formation
- unstable entry into the size box
- poor sheet spreading
- uneven pick-up

Ideal FA Characteristics:

- Equal tension on every beam
- Stable sheet width
- No vibration
- Minimal tension fluctuation
- Smooth acceleration/deceleration
- Fast response to beam diameter changes

How to calculate Weight on warping beam ?

$$W = \frac{L \times N}{Ne_C \times 0.5905}$$

- W = Weight of the warp sheet (g)
- L = Length of the yarn (m)
- Nec = Cotton yarn count (English Cotton Count, Ne)
- 0.5905 = Conversion constant from English cotton count to metric weight
- N = Number of warp ends

We can use the above equation to calculate the weight of the yarn only on the beam , so the total beam weight will be the weight of the empty beam + the weight of the yarn on it which is gradually decreasing as the machine operates .

SO in order to calculate the weight of one single beam we use the above calculations , but in other cases we will need to calculate the weight of 1 meter of a yarn sheet (which contains the total amount of yarn ends that enters single size box).

The weight of 1 meter of a yarn sheet will be used to calculate the tension required in each tension zone including zone **FA** , which is done using the tension table provided by the machine maker .

Each material have its own tension table and there are 3- tables :

- Cotton (page 17)
 - Polyester-cotton (page18)
 - Staple (page 19) which include (polyester spun and viscose yarn)
- File name: Guide to the sizing machine use
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The formula to calculate the tension set point can be found in the excel sheet attached where it has three sheets one for each material .

You just enter the value **X** which refers to the weight of 1 meter of yarn sheet and then it will calculate the required tension set point required for this specific job.

min x	40	min y	240
max x	440	max y	1490
range x	400	range y	1250
x=	180		
x-xmin	140		
xnormalized	0.35	tension	677.5
x normalized=(x-xmin)/(xmax-xmin)			
tension = (Xnormalized *range y)+MinY			

Very important note :

The machine that you will be working on has 2 size box and 20 warping beam position , there are three different operating conditions for the machine .

1. Ten or less warping beams on one size box and the other one not being used .
2. More that Ten warping beams on one size box and the other one not being used .
3. Equal amounts of warping beams on the two size box , the two size box is working at the same time

Note that in the third condition we will calculate the weight of warp sheet that will enter the size box not the total weight of the warp sheet as each size box will have its own load cell roller and proportional pneumatic valve to control the tension of it.

So, we can say that each 10 warping beam positions will be controlled by one Proportional pneumatic valve and in case of using more than 10 warping beams on one size box we will only replicate the output control signal of size box No.1 and apply it on the two Proportional Pneumatic valve .

Component of control circuit :

1. Load cell (Bending beam type)
2. Load cell amplifier
3. Pneumatic proportional valve
4. Plc (controller)